

Module 7. Neural Networks

- ❑ Basic Concepts 
- ❑ Perceptron Learning
- ❑ Artificial Neural Networks
- ❑ Backpropagation
- ❑ Summary

Some slides were developed by Dr. Xuebo Song, who largely borrowed content from George Papadourakis, Prévotet Jean-Christophe; Other slides are from the textbook authors

Introduction

- ❑ Neural networks are a new method of programming computers.
- ❑ They are exceptionally good at performing pattern recognition and other tasks that are very difficult to program using conventional techniques.
- ❑ Programs that employ neural nets are also capable of learning on their own and adapting to changing conditions.
- ❑ Started in early 1940s. Became popular 1980s due to the advances of computer technologies.
- ❑ Much of the inspiration for the field of NNs came from the desire to produce artificial systems capable of sophisticated, perhaps intelligent, computations similar to those that the human brain routinely performs, and thereby possibly to enhance our understanding of the human brain.

Neural Network Techniques

- ❑ Computers have to be explicitly programmed
 - ❑ Analyze the problem to be solved.
 - ❑ Write the code in a programming language.
- ❑ Neural networks learn from examples
 - ❑ No requirement of an explicit description of the problem.
 - ❑ No need for a programmer.
 - ❑ The neural computer adapts itself during a training period, based on examples of similar problems even without a desired solution to each problem. After sufficient training the neural computer is able to relate the problem data to the solutions, inputs to outputs, and it is then able to offer a viable solution to a brand new problem.
 - ❑ Able to generalize or to handle incomplete data.

NNs vs Computers

Digital Computers

- ❑ Deductive Reasoning. We apply known rules to input data to produce output.
- ❑ Computation is centralized, synchronous, and serial.
- ❑ Memory is packetted, literally stored, and location addressable.
- ❑ Not fault tolerant. One transistor goes bad and it no longer works.
- ❑ Exact.
- ❑ Static connectivity.
- ❑ Applicable if well defined rules with precise input data.

Neural Networks

- ❑ Inductive Reasoning. Given input and output data (training examples), we construct the rules.
- ❑ Computation is collective, asynchronous, and parallel.
- ❑ Memory is distributed, internalized, short term and content addressable.
- ❑ Fault tolerant, redundancy, and sharing of responsibilities.
- ❑ Inexact.
- ❑ Dynamic connectivity.
- ❑ Applicable if rules are unknown or complicated, or if data are noisy or partial.

Applications

❑ **classification**

in marketing: consumer spending pattern classification

In defence: radar and sonar image classification

In agriculture & fishing: fruit and catch grading

In medicine: ultrasound and electrocardiogram image classification, EEGs, medical diagnosis

❑ **recognition and identification**

In general computing and telecommunications: speech, vision and handwriting recognition

In finance: signature verification and bank note verification

Applications (Cont.)

❑ **assessment**

In engineering: product inspection monitoring and control

In defence: target tracking

In security: motion detection, surveillance image analysis and fingerprint matching

❑ **forecasting and prediction**

In finance: foreign exchange rate and stock market forecasting

In agriculture: crop yield forecasting

In marketing: sales forecasting

In meteorology: weather prediction

What can you do with an NN and what not?

- ❑ In principle, NNs can compute any computable function, i.e., they can do everything a normal digital computer can do.
- ❑ In practice, NNs are especially useful for **classification** and **function approximation** problems.
- ❑ NNs are difficult to apply successfully to problems that concern manipulation of symbols and memory.
- ❑ There are no methods for training NNs that can magically create information that is not contained in the training data.